

Number Theory Homework

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Exercise 1: Let G be a finite group of order $n \geq 2$, and p a prime such that $p^k \mid n$ for some $k \in \mathbb{N}$. Let $\wp$ be the family of subsets of G having exactly p^k elements. And consider the action of G on $\wp$ as follows:

$$G \times \wp \rightarrow \wp$$

$$(g, X) \mapsto gX = \{gx \mid x \in X\}$$

Question 1.1: Let $X \in \wp$, show that:

1. There is $Y \in O_X$ such that $e \in Y$.
2. O_X contains at most one subgroup of G .
3. O_X contains a subgroup $\Rightarrow \#O_X = n/p^k$

1. Let $x \in X$, since $X \subseteq G$ then $x \in G$ and since it is a group then $x^{-1} \in G$, by considering $g = x^{-1}$ we get that $e = x^{-1}x = gx \in gX$, so $e \in gX \in O_X$.
2. Let H_1, H_2 two subgroups of G with $H_1, H_2 \in O_X$. We have that $H_1 = gH_2$ since they are in the same orbit, H_1 is a subgroup thus $e \in H_1 \Rightarrow e \in gH_2 \Rightarrow g^{-1} \in H_2$, given that H_2 is also a subgroup then $g^{-1}H_2 = H_2 \Rightarrow H_1 = gH_2 = gg^{-1}H_2 = H_2$ so $H_1 = H_2$.
3. Let H be a subgroup in O_X , we have that $S_H = \{g \in G \mid gH = H\} = \{g \in G \mid g \in H\} = H$. $n = \#G = \#O_X \cdot \#S_X = \#O_X \cdot \#S_H = \#O_X \cdot \#H$ therefore, $n = \#O_X \cdot p^k \Rightarrow \#O_X = n/p^k$.

Question 1.2: Suppose that O_X does not contain a subgroup and $Y \in O_X$ such that $e \in Y$.

1. Show that $S_Y \subsetneq Y$ thus $\#S_Y < p^k$.
2. Consider the action $S_Y \times Y \rightarrow Y, (g, y) \mapsto gy$. Prove that $\forall y \in Y, \#O_y = \#S_Y$, and deduce that $\#(S_Y) = p^l$ for some $l \in \llbracket 0, k-1 \rrbracket$.

1. We have $S_Y = \{g \in G \mid gY = Y\}$ since $e \in Y$ then $Y = \{g \in G \mid g \in Y\} \subseteq \{g \in G \mid gY = Y\} = S_Y$, and since S_Y is a subgroup then $Y \neq S_Y$ therefore $S_Y \subsetneq Y$. Since S_X and S_Y are conjugates then $\#S_X = \#S_Y$ hence $\#S_X = \#S_Y < \#Y = p^k$.
2. Let $y \in Y$, we have that $O_y = \{gy \mid g \in S_Y\} = S_Y y$ and thus $\#O_y = \#S_Y y = \#S_Y$. By the class equation, we have that $\#Y = \sum_{i=1}^r \#O_{y_i} = r \#O_y = r \#S_Y$ and since $\#Y = p^k$ then $\#S_Y \mid p^k$ therefore we get $\#S_Y = p^l$ for $l \in \llbracket 0, k-1 \rrbracket$.

Question 1.3: Let $v \geq 0$ be the number of subgroups of G with order p^k , $l_1, \dots, l_w \in \llbracket 0, k-1 \rrbracket$ and w the number of orbits O_X which cannot contain a subgroup.

$$\#\wp = v \frac{n}{p^k} + \left(\frac{n}{p^{l_1}} + \dots + \frac{n}{p^{l_w}} \right)$$

Consider the distinct orbits $O_{x_1}, \dots, O_{x_v}, O_{x_{v+1}}, \dots, O_{x_{v+w}}$ where $O_{x_1}, \dots, O_{x_v}$ are the orbits that have subgroups, while $O_{x_{v+1}}, \dots, O_{x_{v+w}}$ are the orbits that do not, we have

$$\begin{aligned} \#\wp &= \sum_{i=1}^v \#O_{x_i} + \sum_{i=v+1}^{v+w} \#O_{x_i} \\ &= \sum_{i=1}^v p^k + \sum_{i=v+1}^{v+w} \frac{\#G}{\#S_{x_i}} \end{aligned}$$

from A.3, we have that $\#O_{x_i} = n/p^k$ and from B.2, $\#S_{x_i} = p^{l_i}$, therefore

$$\#\wp = v \frac{n}{p^k} + \sum_{i=1}^w \frac{n}{p^{l_i}}$$

Question 1.4:

1. Rewrite the last equality for a cyclic group G .
2. Deduce that

$$v \frac{n}{p^k} + \sum_{i=1}^w \frac{n}{p^{l_i}} = \frac{n}{p^k} + \sum_{i=1}^z \frac{n}{p^{m_i}}$$

3. Conclude that $v \equiv 1 \pmod{p}$.

1. Suppose G is a cyclic group, then there exists a unique subgroup of order p^k in G , so $v = 1$ and thus we obtain in the new group that

$$\#\wp' = \frac{n}{p^k} + \sum_{i=1}^z \frac{n}{p^{m_i}}$$

2. Given that $\#\wp$ only depends on the cardinal of G , thus $\#\wp = \#\wp'$ since G as a cyclic group or not have the same cardinality, therefore

$$v \frac{n}{p^k} + \sum_{i=1}^w \frac{n}{p^{l_i}} = \frac{n}{p^k} + \sum_{i=1}^z \frac{n}{p^{m_i}}$$

3. We have

$$\begin{aligned} v \frac{n}{p^k} + \sum_{i=1}^w \frac{n}{p^{l_i}} &= \frac{n}{p^k} + \sum_{i=1}^z \frac{n}{p^{m_i}} \\ \Rightarrow (v-1) \frac{n}{p^k} &= \sum_{i=1}^z \frac{n}{p^{m_i}} - \sum_{i=1}^w \frac{n}{p^{l_i}} \\ \Rightarrow v-1 &= \sum_{i=1}^z p^{k-m_i} - \sum_{i=1}^w p^{k-l_i} \end{aligned}$$

since m_i and l_i are strictly smaller than k then $p \mid \sum_{i=1}^z p^{k-m_i} - \sum_{i=1}^w p^{k-l_i}$ therefore $p \mid v-1 \Rightarrow v \equiv 1 \pmod{p}$.

